Master 104 – Continuous Time Finance – Exam - October, 24th 2025.

No documents allowed. No electronic device of any kind allowed (cell phone, electronic calculator...). The exam lasts 2 hours. You can write your answers in French or in English. **For each question, recall the number of the question and write only the result. Do not write any justifications nor intermediate computations. Do not leave drafts on your paper.**

In all the exam $\mathbb{V}[X]$ denotes the variance of X, (B_t) is a standard Brownian motion under $\mathbb{P}$ and $x^+ = \max(0, x)$.

Consider the binomial market model with time horizon $T = 1$, $N = 2$ periods of time, probability $p = 0.2$ that the price moves up with a rate $u = 1.2$ and $1 - p$ that it moves down with a rate $d = 0.8$. The risk–free rate $r = 0$ and the initial risky asset price $S_0 = 100$. Consider the options payoff $\phi(S_0, \ldots, S_T) = (S_T - K)^+$ where $K = 110$. Give **Q1**: the value of the risk-neutral probability denoted q, **Q2**: the price denoted v_0 of the claim ϕ, **Q3**: the prices v_u and v_d of the claim at time t_1, **Q4**: the prices v_{uu} and v_{ud} and v_{dd} of the claim at time t_2, **Q5**: the delta of the claim Δ_0 at time $t = 0$, **Q6**: the deltas of the claim Δ_u and Δ_d at time $t = 1$, **Q7**: the amount of cash held in the strategy portfolio c_0 at initial time, **Q8**: the amount of cash the amounts of cash c_u and c_d in the strategy portfolio at time t_1.

Give **Q9**: $\mathbb{E}[B_t]$, **Q10**: $\mathbb{E}[B_t^2]$, **Q11**: $\langle B \rangle_t$, **Q12**: $\mathbb{E}[B_t B_{t+s}]$ with $s > 0$, **Q13**: $\mathbb{V}[B_s + B_t]$.

Consider $Z_T := e^{\sigma B_T - \frac{1}{2}\sigma^2 T}$. Give **Q14**: $\mathbb{E}[Z_T]$, **Q15**: $\mathbb{E}[Z_T B_T]$. **Q16**: $\mathbb{E}[Z_T^2]$. **Q17**: $\mathbb{V}[Z_T]$.

Consider the stochastic differential equation $\mathrm{d}S_t = \kappa(b - \ln S_t)S_t \mathrm{d}t + \sigma S_t \mathrm{d}B_t$, with $S_0 = s_0 > 0$, $\kappa > 0$, $b > 0$ and $\sigma > 0$. Let $Y_t = \ln(S_t)$. Give **Q18**: $\mathrm{d}Y_t$, **Q19**: S_t, **Q20**: $\mathbb{E}[S_t]$, **Q21**: $\lim_{t \to +\infty} \mathbb{E}[S_t]$. **Q22**: $\mathbb{V}[S_t]$.

Consider $X_t = e^{\alpha B_t} \int_0^t e^{\gamma B_s} ds$ where α, γ are two real numbers. Give **Q23**: $\mathrm{d}X_t$, **Q24**: $\mathbb{E}\big[X_t\big]$.

Consider $\mathrm{d}X_t = \kappa(B_t - X_t)\mathrm{d}t + \sigma \mathrm{d}B_t$, $X_0 = x$, and $\kappa, \sigma > 0$. Denote $Y_t = e^{\kappa t} X_t$. Give **Q25**: $\mathrm{d}Y_t$, **Q26**: X_t, **Q27**: $\mathbb{E}[X_t]$, **Q28**: $\mathbb{V}[X_t]$, **Q29**: $\lim_{t \to +\infty} \mathbb{V}[X_t]$.

Consider $X_t = e^{-\kappa t} \int_0^t \sigma e^{\kappa s} \mathrm{d}B_s$. **Q30**: Give $\mathrm{d}X_t$. Let $Y_t = e^{X_t}$. **Q31**: Give $\mathrm{d}Y_t$. Let $F(t, T)$ the futures price at time t for delivery at T. Consider $\mathrm{d}F(t, T) = \sigma e^{-\kappa(T-t)} \mathrm{d}B_t$, with $t \in (0, T)$ and initial condition $F(0, T) = f(T) > 0$ for all $T \geq 0$, f differentiable. **Q32**: Give $\mathbb{E}[F(t, T)]$, **Q33**: Give $F(t, T)$. Let $S_t = F(t, t)$. Give **Q34**: $\mathrm{d}S_t$, **Q35**: $\lim_{t \to +\infty} \mathbb{E}[S_t]$.

Consider the maximisation of expected utility of the terminal wealth $\sup_\theta \mathbb{E}\big[U(X_T)\big]$ where $U(x) := x^\gamma$ with $\gamma \in (0, 1)$ and $\mathrm{d}X_t = [\mu\theta_t + r(1 - \theta_t)]X_t \mathrm{d}t + \sigma\theta_t X_t \mathrm{d}B_t$ with $\mu > r > 0$ and $\sigma > 0$ and $X_0 = x > 0$. **Q36**: Write the HJB equation satisfied by the value function $v(t, x)$ of this problem. Assume $v(t, x) = f(t)x^\gamma$. **Q37**: Give the ODE satisfied by f. **Q38**: Give $f(t)$. **Q39**: Give the optimal trading strategy θ^*. **Q40**: Give $x^* \in \mathbb{R}$ such that $U(x^*) = \mathbb{E}[U(X_T^{\theta^*})]$.

Solutions

Q1. 0.5.
Q2. 8.5.
Q3. $v_u = 17$, $v_d = 0$.
Q4. $v_{uu} = 34, v_{ud} = 0, v_{dd} = 0$.
Q5. $\frac{17}{40}$.
Q6. $\Delta_u = 17/24$ and $\Delta_d = 0$.
Q7. -34.
Q8. $c_u = -68$ and $c_d = 0$.
Q9. $\mathbb{E}[B_t] = 0$.
Q10. $\mathbb{E}[B_t^2] = t$.
Q11. Give $\langle B \rangle_t = t$.
Q12. Give $\mathbb{E}[B_t B_{t+s}]$ with $s > 0 = \min(t, t+s) = t$.
Q13. Give $\mathbb{V}[B_s + B_t] = \mathbb{V}[B_s] + \mathbb{V}[B_t] + 2\mathrm{Cov}[B_s, B_t] = t + s + 2\min(t, s)$.
Q14. Give $\mathbb{E}[Z_T] = 1$.
Q15. Give $\mathbb{E}[Z_T B_T] = \sigma T$.
Q16. Give $\mathbb{E}[Z_T^2] = e^{T\sigma^2}$.
Q17. Give $\mathbb{V}[Z_T] = e^{T\sigma^2} - 1$.
Q18. Give $dY_t = (k(b - Y_t) - \frac{1}{2}\sigma^2)dt + \sigma dB_t, \quad Y_t = \ln(S_t)$.
Q19. Give $S_t = e^{\left(\ln(S_0)e^{-\kappa t} + (b - \frac{\sigma^2}{2\kappa})(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)} dB_s\right)}$.
Q20. Give $\mathbb{E}[S_t] = e^{\left(\ln S_0 e^{-\kappa t} + (b - \frac{\sigma^2}{2\kappa})(1 - e^{-\kappa t}) + \frac{1}{2}\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})\right)}$.
Q21. Give $\lim_{t \to +\infty} \mathbb{E}[S_t] = e^{(b - \frac{\sigma^2}{4\kappa})}$.
Q22. Give $\mathbb{V}[S_t]$. We can use the fact that $S_t = e^{Y_t}$ is a log-normal random variable and we have the general formula that $\mathbb{V}(S_t) = \mathbb{E}[S_t]^2(e^{(\sigma_y^2)} - 1)$ where $\sigma_y^2 = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$. We can use the result for $\mathbb{E}[S_t]$.
Q23. Give $dX_t = [e^{B_t(\gamma+\alpha)} + \frac{1}{2}\alpha^2 X_t]dt + \alpha X_t dB_t$.
Q24. Give $\mathbb{E}[X_t] = \frac{e^{(\frac{1}{2}\alpha^2 + C)t} - e^{\frac{1}{2}\alpha^2 t}}{C}$ where $C = \alpha\gamma + \frac{1}{2}\gamma^2 = \frac{1}{2}\gamma(2\alpha + \gamma)$.
Q25. Give $dY_t = e^{\kappa t}\kappa B_t dt + \sigma e^{\kappa t} dB_t$.
Q26. Give $X_t = e^{-\kappa t}x + \kappa \int_0^t e^{-\kappa(t-s)} B_s ds + \sigma \int_0^t e^{-\kappa(t-s)} dB_s$ or also

$$X_t = e^{-\kappa t}x + B_t + (\sigma - 1)\int_0^t e^{-\kappa(t-s)} dB_s,$$

using Ito on the expression $e^{-\kappa s}B_s$.
Q27. Give $\mathbb{E}[X_t] = e^{-\kappa t}x$.
Q28. Give $\mathbb{V}[X_t] = t + \frac{(\sigma-1)^2}{2\kappa}(1 - e^{-2kt}) + \frac{2(\sigma-1)}{\kappa}(1 - e^{-\kappa t})$.
Q29. Give $\lim_{t \to +\infty} \mathbb{V}[X_t] = \infty$.
Q30. Give $dX_t = -\kappa X_t dt + \sigma dB_t$.
Q31. Give $dY_t = Y_t(-\kappa X_t + \frac{1}{2}\sigma^2)dt + \sigma Y_t dB_t$ with $Y_t = e^{X_t}$.
Q32. Give $\mathbb{E}[F(t, T)] = f(T)$.
Q33. Give $F(t, T) = f(T) + \int_0^t \sigma e^{-\kappa(T-s)dB_s}$.
Q34. Give $dS_t = (f'(t) + \kappa f(t) - \kappa S_t)dt + \sigma dB_t$.
Q35. Give $\lim_{t \to +\infty} \mathbb{E}[S_t] = \lim_{t \to \infty} f(t)$.

Q36. Write the HJB equation satisfied by the value function $v(t,x)$ of this problem. The HJB is given by

$$\begin{cases} -\partial_t v(t,x) - \sup_\theta \left\{ \partial_x v(t,x)(\mu\theta + r(1-\theta))x + \partial_{xx} v(t,x) \frac{1}{2}\sigma^2\theta^2 x^2 \right\} = 0, \\ V(T,x) = x^\gamma. \end{cases}$$

Q37. Give the ODE satisfied by f. The ODE sastifies for the function f(t) is given by:

$$f'(t) + f(t)\rho = 0,$$

where the expression ρ is defined by:

$$\rho = \gamma \sup_\theta \left\{ (\mu\theta + r(1-\theta)) + (\gamma - 1)\frac{1}{2}\sigma^2\theta^2 \right\}.$$

Q38. The function $f(t)$ is given:

$$\begin{cases} f(t) = e^{\rho(T-t)}, \\ f(T) = 1. \end{cases}$$

Q39. Give the optimal trading strategy θ^*.

$$\theta^* = \left(\frac{\mu - r}{\sigma^2(1-\gamma)} \right)$$

Q40. Give $x^* \in \mathbb{R}$ such that $U(x^*) = \mathbb{E}[U(X_T^{\theta^*})]$.

$$U(x^*) = \mathbb{E}[U(X_T^{\theta^*})] = \sup_\theta \mathbb{E}[U(X_T^\theta)] = v(0,x) = f(t)x^\gamma$$

That is

$$x^\gamma = e^{\rho(T)} x^\gamma \Rightarrow x^* = e^{\frac{\rho}{\gamma}(T)} x$$